# LAB / TIMERS

## 1. OVERVIEW

### 1.1 Goal

The aim of this laboratory is to put into practice the use of timers, watchdog, interrupts, PWM.

## 1. TASK ONE

In this first part, we want to blink a LED (GREEN_LED) at a frequency of 1 Hz (duty of 50%).

- Use the timer TIM1 to do this job.
- Use of interruptions will be mandatory.
- The main program will initialize the timer and interruptions. Then the forever loop will do nothing.

To use the timer, you must know how it works (explained in course). To remember, you must start the timer in interrupt mode (`HAL_TIM_Base_Start_IT()`) and you must implement the callback interrupt routine `HAL_TIM_PeriodElapsedCallback`.

Caution, you must activate “TIM1_Update interrupt” in CubeMX.

## 2. TASK TWO

In this second part, we want to measure an external frequency between 100 Hz and 100kHz at a 1Hz resolution. The signal to measure will be injected by “Analog Discovery 2”.

To measure the external signal, we will use the timer TIM2 to have the processor input PA5 as clock source for timer. As already used for the interrupt laboratory, you must

connect the AnalogDiscovery to the pin 3 of 26 pin connector to inject the signal. The pin 24 is the ground.

Select the clock source as ETR2.

To remember, you must start the timer (`HAL_TIM_Base_Start()`). To read the timer value, use `__HAL_TIM_GET_COUNTER()` and to write it use `_HAL_TIM_SET_COUNTER()`. Use SysTick to know when one second is elapsed.

The goal is to send this frequency to the serial debug port.

If we want to measure a signal between 1 and 100Hz with a resolution of 100mHz, what will be the best solution and how to implement it?

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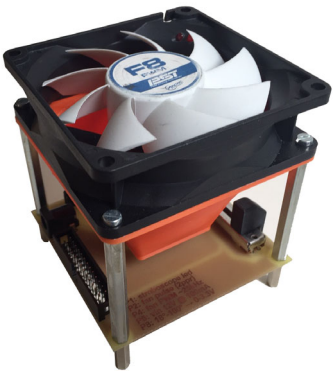
# 1. TASK THREE

## 1.1 Goal

Now we want to control the speed of a PC fan using a PWM (Pulse Width Modulation) signal.

## 1.2 Hes-so – fan module

The laboratory is equipped with a fan with a lot of functionality. This chapter will describe how it works.



Sin - fan

A potentiometer allows you to heat a resistor placed in the air cone. We will not use it in this laboratory. Leave it in the far-left position.

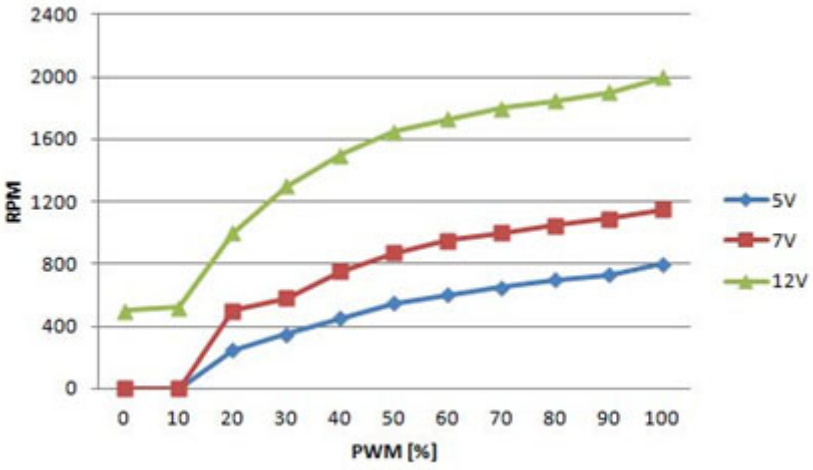
A DC-B connector must be connected to a voltage **between 5 to 12 volts** depending on the fan speed you want.



A 26-pin connector provides you with the following signals:

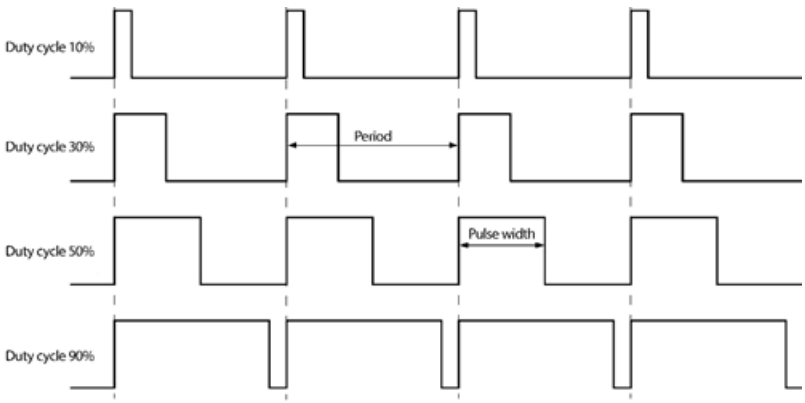
| Pin / Wire | Function |
|---|---|
| **P1** | Led stroboscope |
| **P2** | Sense (2 pulses per revolution) |
| **P4** | PWM control signal |
| **P8** | Temperature (analogue output) |
| **P24** | GND |
| **P26** | VCC |

To operate, the fan requires a PWM frequency between 21 and 25 kHz. The fan speed depends on the duty cycle of this PWM as shown in the picture below.




## 1.3 The PWM (Pulse Width Modulation)

A pulse-width modulated signal is a signal whose frequency is fixed but whose duty cycle can change.

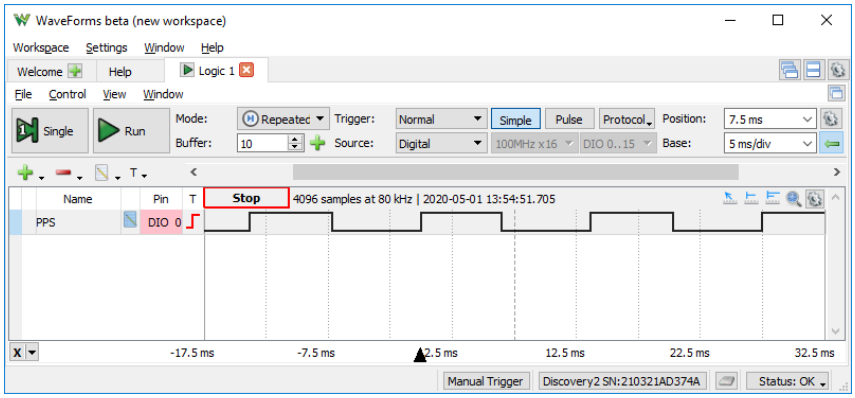


Our microcontroller has a dedicated module to perform this task.

## 1.4 PWM configuration

The timer TIM2 will be used to generate this PWM signal on the “Channel 1”. You must configure it to have a frequency of 25kHz. Set the PWM output to the pin PA5, so this will correspond to the extension board connector pin 3.

To start the timer in PWM mode you must call `HAL_TIM_PWM_Start()`. To change the duty cycle, you just need to write in the compare register the value comprise between 0 and the counter period you have chosen (`TIM2->CCR1 = xxx`). Control you PWM is working with the AnalogDiscovery. Then, you can connect it to the fan.



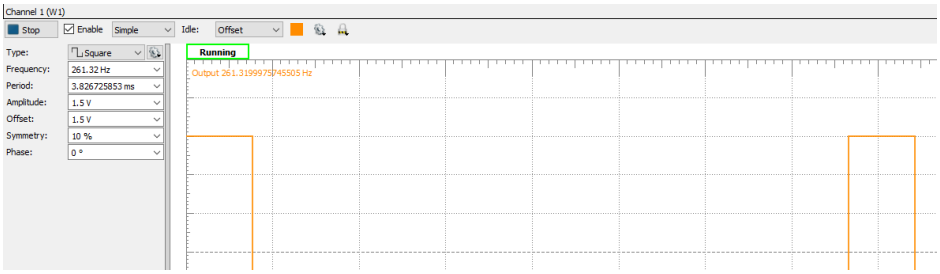

What is the resolution of your PWM? .....

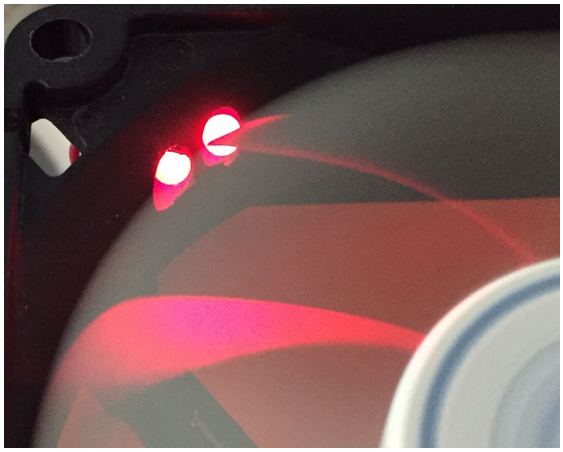
## 1.5 Tests and measure

To measure the fan speed, we will measure the signal “Sense”. This signal will be connected to the pin 13 of the extension board (PG0). This signal will be configured as interrupt input. To measure the time between interrupts, we will configure timer TIM1 at an appropriate speed.

For each interrupt (rising edge for example), we will measure the TIM1 counter to know the speed of the fan.

Using the “Wavegen” output of the "Analog Discovery 2" analyzer, generate a signal on pin 1 of the fan (strobe led). This signal will allow you to see the exact speed of the fan. Be careful, it is necessary to generate a frequency 9 times higher than the fan rotation frequency because the fan has 9 blades. Select a small duty cycle as the example below to have a nice stroboscope effect.



## 1.6 Regulation

By measuring the time between two "Sense" signal to know the speed of your fan create a simple PWM regulator to set the speed at 11.111Hz.

Why do we choose this special frequency ? ……

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If you choose a duty cycle of 0%, what happens and why? ……

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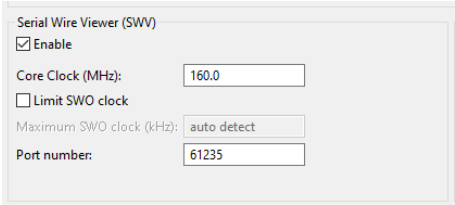
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## 1.7 Serial Wire Viewer (SWV)

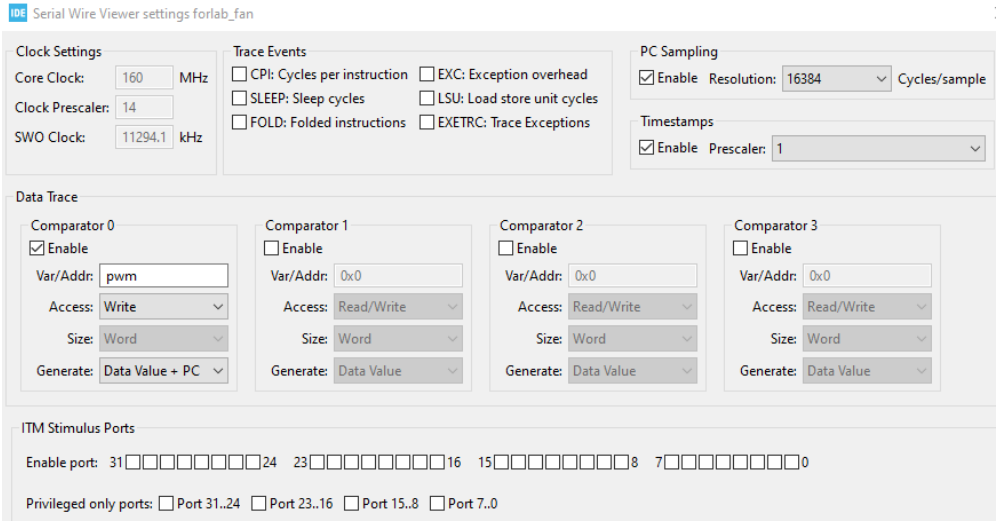
An incredible connection offers you the ability to see graphically a variable, for example. You can see the description below to activate this utility.

Enable SWV in debugger (debug configuration…):

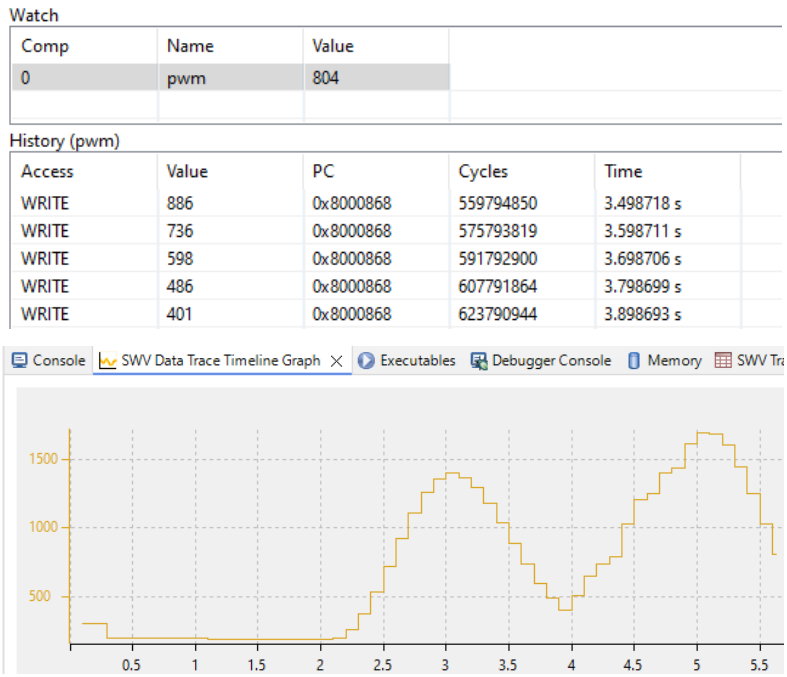



In debug mode, open the window below (Windows - Show view – SWV – SWV data trace)

With the tab “Configure Trace”, select configuration as below:




You can now see the pictures below in SWV data trace and in SWV data Trace Timeline:



## 1.8 STM32CubeMonitor

With the usage of STM32CubeMonitor, you can see some variables in real time in a graphical interface. The example below shows you the result of changing the fan voltage.

This is another software, not STM32CubeIDE.

The usage is not tricky but remember that it could be useful for a project.

